

ABA ERP Project - Work Plan

Corrected development sequence and project touchpoints

1. Project Objective

The goal is to develop a lightweight ERP application for managing the client's business operations across customers, referrals, commissions, sales, payments, procurement, production, inventory, finance, POS, and reporting. The platform will operate as one central application with connected modules so information can flow across business functions without repeated data entry.

2. Agreed Delivery Direction

The development sequence will follow the order already agreed with the technical team. The first build priority is the customer-facing foundation, followed by the multi-level referral structure and commission engine. Procurement, back-order, inventory, and manufacturing features will be introduced after these core commercial modules are established.

3. Immediate Project Touchpoints

The following touchpoints reflect the near-term commitments already discussed and should be used to track what stakeholders can expect before full development progresses.

Touchpoint	Expected Output
Customer call - tomorrow	Clarify outstanding requirements, confirm immediate client expectations, and capture questions to validate during the ABA visit.
Work plan submission - Friday	Submit the corrected work plan showing the agreed build order, phase outputs, dependencies, and immediate touchpoints.
Yusuf design steps - 10 September 2026	Provide the initial design direction, application flow, priority screens, and design actions needed for the first modules.
ABA client visit - next week	Observe the client's actual business operations and validate production, inventory, sales, merchant/POS, finance, referral, and payment processes.
Post-visit requirements review	Consolidate findings from the visit and client discussions into a confirmed business-process and system-requirement document.
Design and architecture confirmation	Confirm workflows, database structure, roles, permissions, module boundaries, API approach, security requirements, and MVP scope.
Development commencement	Begin with Customer Module -> Multi-Level Referral -> Commission Engine before moving into later ERP modules.

4. Proposed Technology Stack

Area	Technology / Approach
Backend	NestJS; TypeScript; REST API architecture
Database	PostgreSQL as the central relational database for operational and financial data
Frontend	Modern responsive web application for desktop, tablet, and mobile
Infrastructure	Amazon Web Services (AWS) for hosting, managed database infrastructure, storage, backups, monitoring, logging, security, and scalability

5. System Modules

1. Authentication and User Management	9. Commission Management
2. Customer Management	10. Payments and Payouts
3. Product Management	11. Finance and Accounting
4. Merchant Management	12. Procurement
5. Supplier Management	13. Inventory Management
6. Sales and Order Management	14. Production Management
7. Merchant POS	15. Reporting and Dashboard
8. Referral Management	16. Audit and Activity Tracking

Additional integrations such as logistics, escrow payments, hardware POS integration, and advanced analytics can be introduced progressively after the core platform is stable.

6. Development Work Plan

Phase 1 - Requirements & Process Review

Key activities:

- Review existing requirements and prior decisions
- Conduct customer call and ABA client visit
- Confirm customer, sales, merchant/POS, referral, commission, payment, finance, production, procurement, inventory, reporting, and integration requirements
- Identify gaps, dependencies, approval points, and items that must be confirmed before build

Deliverable: Confirmed Business Process and System Requirements Document

Phase 2 - System & UI/UX Design

Key activities:

- Define application architecture and modular NestJS structure
- Design PostgreSQL database model and major entity relationships
- Define API structure, roles, permissions, audit and security requirements
- Prepare user flows, wireframes, dashboard direction, and priority screens
- Confirm MVP scope and build sequence

Deliverable: Approved Architecture, Database Design, API Structure, Wireframes and MVP Scope

Phase 3 - Customer Module & Core Foundation

Key activities:

- Set up authentication and role management
- Develop customer registration and profile management
- Create core product/catalogue entities required by the customer flow
- Establish shared entities and audit foundations needed by later modules

Deliverable: Working Customer Module and Core Platform Foundation

Phase 4 - Multi-Level Referral System

Key activities:

- Capture referral codes and referral relationships
- Maintain explicit upline/downline hierarchy
- Support configured referral levels
- Provide network views and relationship traceability
- Prepare referral events for commission processing

Deliverable: Working Multi-Level Referral Module

Phase 5 - Commission Engine

Key activities:

- Define configurable commission policies and eligible levels
- Resolve upline chain for qualifying purchases
- Calculate commissions and create ledger entries
- Support pending, approved, paid, and reversed commission states
- Maintain full traceability to originating transactions

Deliverable: Working Commission Engine and Commission Ledger

Phase 6 - Sales, Orders & Basic POS

Key activities:

- Develop standard order and sales workflows
- Connect customers, products, referrals, and qualifying purchases
- Implement merchant sales and basic POS flow
- Record payment method and sales status
- Ensure sales events feed commission and reporting processes

Deliverable: Sales, Order and Basic POS Modules

Phase 7 - Payments, Payouts & Finance Foundation

Key activities:

- Implement basic payment tracking
- Support commission payout workflow
- Introduce merchant settlement where confirmed
- Set up finance foundations for receivables, payables, expenses, and transaction traceability
- Introduce payment-provider or escrow integration only where requirements are confirmed

Deliverable: Payment, Payout and Finance Foundation

Phase 8 - Procurement, Inventory & Production

Key activities:

- Develop supplier and procurement workflows

- Implement purchase requisitions, purchase orders, and goods received
- Track raw materials, work-in-progress, and finished goods
- Implement production orders, material allocation, costing, and wastage tracking
- Introduce back-order handling only after the confirmed business process is available

Deliverable: Integrated Procurement, Inventory and Production Modules

Phase 9 - Reporting & Analytics

Key activities:

- Develop management dashboard and operational reporting
- Provide sales, referral, commission, inventory, production, and finance reporting
- Add agreed management KPIs and audit/activity visibility

Deliverable: Reporting and Management Dashboard

Phase 10 - Integration & Harmonisation

Key activities:

- Connect module workflows end-to-end
- Validate data movement across customer, referral, commission, sales, finance, inventory, and production
- Remove duplicate entry points and resolve cross-module inconsistencies

Deliverable: Harmonised Integrated ERP Platform

Phase 11 - Testing & UAT

Key activities:

- Functional and API testing
- User-role and permission testing
- Referral and commission calculation testing
- Sales, payment, inventory, and production workflow testing
- Security and performance testing
- Client acceptance testing and issue resolution

Deliverable: UAT Sign-off and Release Candidate

Phase 12 - Deployment & Support

Key activities:

- Production deployment and database setup
- User and admin onboarding
- Merchant onboarding where applicable
- Training and documentation
- Monitoring, bug fixes, and early performance improvements

Deliverable: Live Production System and Handover

7. Development Approach

The platform will use a modular monolith architecture with NestJS and PostgreSQL, deployed on AWS. Each business function will remain separated into its own module while sharing the required entities and transactions across the platform. The priority is to establish the customer, referral, and commission flow first, then expand into sales, finance, procurement, inventory, and manufacturing. This keeps the early build aligned with the agreed commercial workflow while allowing the ERP functions to be added without rebuilding the core application.

8. Immediate Next Steps

1. Complete the customer call and capture unresolved requirements.
2. Submit the corrected work plan on the agreed Friday deadline.
3. Receive Yusuf's design steps by 10 September 2026.
4. Conduct the ABA client visit next week and observe the current operating process.
5. Consolidate visit findings and confirm the MVP requirements.
6. Approve system architecture, database design, workflows, and UI direction.
7. Begin development with the Customer Module, then Multi-Level Referral, then the Commission Engine.
8. Defer procurement, inventory, production, and back-order implementation until their business processes are validated and the earlier modules are established.

Sequence note: Customer Module -> Multi-Level Referral -> Commission Engine is the committed first development path. Procurement, inventory, production, and back-order features are later-stage work and must not be shown ahead of the referral and commission build.